PENNSYLVANIA RAILROAD SYSTEM

Locomotive Maintenance Instructions No. L-5-C

ISSUED ALTOONA, PA.
DECEMBER 18, 1923.

Method of Applying and Maintaining Flues and Arch Pipes in Boilers and Instructions for Welding Flues to Flue Sheets

(SUPERSEDING LOCOMOTIVE MAINTENANCE INSTRUCTIONS NO. L-5-B, DATED JUNE 15, 1918.)

GENERAL

1. **Flues and Arch Pipes:** Boiler flues and arch pipes must be in accordance with Standard Specification No. 17 and the tracings designated.

Where specifications, instructions or tracings are referred to, the latest issues of the same are intended.

2. **Safe Ends:** Safe ending material must be .134 inch in thickness, B. W. G. No. 10, for repairing all flues or tubes that have a wall thickness of .134 or less. For repairing all flues or tubes that have a wall thickness of .148 inch, B. W. G. No. 9, or over, the thickness of the safe ending material must be the same as that of the flue repaired.

The outside diameter of safe ending material must be the same as that of the flue or tube ends to which it will be applied.

Where possible, the safe ends should be applied to the fire box end of the flues. When the length of the swaged ends of large superheater flues is 16", the safe ends must be applied to the smoke box end of the flues.

3. **Ferrules:** Ferrules for boiler flues, including those for superheater flues, must be made of soft copper, which is .075 inch thick. The length of the ferrules must be $\frac{1}{32}$ inch greater than the thickness of the flue sheeet. The outside diameter of the ferrules must conform to dimensions shown on standard tracings. Where the outside diameter is not shown on any tracing, it should be made the same as the outside diameter of the flue.

4. **Preparation of Fire Box Sheets:** The diameter of holes in fire box sheets for flues and arch pipes must conform to dimensions shown on standard tracings. Where tracings do not show dimensions, the diameter of the holes must be the same as the outside diameter of the flues and arch pipes.

All holes in fire box sheets for flues and arch pipes must have the inside and outside edges rounded to $\frac{1}{16}$" radius as indicated by Sketch "A".



SKETCH "A"